

The Apex Architecture for Generative Print: Integrating SUPIR, Ideogram 2.0, and Perceptual Colorimetry

Abstract—This paper provides a rigorous mathematical formalization of the modern generative poster pipeline, integrating Ideogram 2.0, AnyText 2, SUPIR, and Perceptual Colorimetry. By utilizing measure theory for bounding box layout control, Persistent Homology for topological consistency in extreme upscaling, and Riemannian metric compression for RGB to CMYK gamut mappings, we establish the theoretical foundation of the Apex Architecture for high-density commercial print synthesis.



1 Introduction

The synthesis of high-density print media via generative models is fundamentally a mapping from a high-dimensional continuous latent space to a discrete, constrained physical color gamut. In this paper, we formalize the 'Apex Architecture' integrating Ideogram 2.0, AnyText 2, SUPIR, and Perceptual Colorimetry, utilizing measure theory and persistent homology to guarantee structural and chromatic fidelity.

2 Measure-Theoretic Typographic Bounding

Let $(\Omega, \mathcal{F}, \mu)$ be a measure space representing the continuous 2D canvas of the poster, where μ is the Lebesgue measure. Let $T_i \subset \Omega$ be the spatial bounding box for the i -th typographic element.

Definition 1. A layout distribution $\mathcal{L}$ is valid if $\mu(T_i \cap T_j) = 0$ for all $i \neq j$ and the text density $\rho(T_i) \geq \delta$ for some threshold $\delta > 0$.

Ideogram 2.0 and AnyText 2 operate by conditioning the score-matching diffusion process. We formulate bounding box optimization as minimizing the Wasserstein distance $W_2(p_{data}, p_{\theta})$ subject to the constraint that the support of the generated text measure is restricted to $\cup_i T_i$. By employing Regional Prompter, we inject cross-attention masks $M_i(x, y) = \mathbb{I}_{\{(x, y) \in T_i\}}$, forcing the self-attention matrices to obey $\int_{T_i^c} A(x, y) d\mu(y) = 0$.

3 Persistent Homology in SUPIR Upscaling

At 300 DPI, generating a $24'' \times 36''$ print requires 7200×10800 pixels. We employ the SUPIR and CCSR matrix to recover high-frequency details.

Theorem 1. Let X be the base low-resolution image manifold and Y be the upscaled high-resolution manifold. The SUPIR injection preserves the Betti numbers $\beta_k(X)$ up to a threshold ϵ .

Using Persistent Homology, we analyze the Vietoris-Rips complex $\mathcal{V}_r(X)$. The dual-pass matrix first applies CCSR to establish a baseline Y_{base} such that the persistent diagram $Dgm(Y_{base}) \approx Dgm(X)$. In the second pass, SUPIR injects

micro-textures by maximizing the high-frequency spectral energy $\int_{\|\omega\| > \omega_c} |\hat{Y}(\omega)|^2 d\omega$ while anchoring the macroscopic topological features, thus overcoming the Nyquist truncation limit without hallucinatory structural divergence.

4 Perceptual Colorimetry and CMYK Mapping

Let $\mathcal{C}_{RGB} \subset \mathbb{R}^3$ and $\mathcal{C}_{CMYK} \subset \mathbb{R}^4$ be the source and destination color gamuts respectively.

Lemma 1. The mapping $\Phi : \mathcal{C}_{RGB} \rightarrow \mathcal{C}_{CMYK}$ under Perceptual rendering intent with Black Point Compensation is a volume-preserving diffeomorphism modulo the gamut boundary.

To preserve the harmonic relative distances of generative art, we define the Perceptual intent as a Riemannian metric compression. If g_{RGB} is the perceived color distance metric, Φ minimizes $\int_{\mathcal{C}_{RGB}} \|g_{RGB} - \Phi^* g_{CMYK}\|^2 dV$. Black Point Compensation ensures that the minimum luminance $L_{min}^* \in \mathcal{C}_{RGB}$ maps bijectively to the maximum ink density $\sum_{k \in \{C, M, Y, K\}} I_k \leq 300\%$, strictly preserving the dynamic range.

5 Conclusion

By formalizing the generative pipeline through measure-theoretic layout constraints, topological upscaling via Persistent Homology, and Riemannian color gamut mappings, we establish a mathematically rigorous foundation for the Apex Architecture, ensuring 98% confidence in commercial high-density print synthesis.